KNAUF FIREBOARD

Functional Requirements Specification

VERSION 1.1 • ENGLISH

Version: 1.1

Status: Finalized functional requirements – production acceptance pending

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Prototype handover: Knauf Digital receives a ZIP copy of the prototype source code, including the referenced data, tests and documentation.

Purpose: Functional requirements for the production application; suitable for adaptation into Confluence.

Revision source: docs/fireboard-requirements-specification.md (v1.0); approved product-owner decisions applied in v1.1. The original 57 requirement IDs are preserved and DR-012–DR-016 are added.

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1. Introduction and Purpose

Executive Summary

Fireboard determines the required fire-protection thickness for supported steel profiles and provides indicative material estimates and project documentation. The existing prototype is the functional reference for the production application, including its calculation data, supported configurations and demonstrated workflows.

The production application must support standard and custom profiles, the four-step calculation workflow, project editing and local save/reopen, material and environmental information, Danish/English presentation, and the approved PDF, Excel, ZIP and editable project-file exports. This specification separates those requirements from prototype limitations and records acceptance criteria and outstanding decisions for formal handover review through Confluence. Knauf Digital is responsible for selecting and implementing the production technology, architecture and operational arrangements.

Fireboard supports the selection of Knauf Fireboard fire-protection thickness for steel profiles. It combines profile and fire-design inputs with calculation data to produce dimensioning results, indicative material consumption, product environmental information and project documentation.

This specification describes **what the production application must do**. The existing application is a functional prototype and reference implementation. Knauf Digital is responsible for technical implementation, architecture, hosting, security, maintenance and operation.

The requirements draw on the functional and product-scope decisions supplied by Knauf Commercial / the designated product owner, the existing as-built functional specification (docs/functional-specification.md), and the source, data and tests in the prototype ZIP supplied to Knauf Digital. Approval of functional scope does not constitute engineering, regulatory or environmental approval; the relevant responsibility categories are distinguished in section 12. The as-built document describes an earlier baseline and has not been changed. Where it differs from the current code or the approved decisions, this document identifies the distinction rather than carrying obsolete statements into the target requirements.

1.1 Requirement conventions

Prefix	Meaning
FR	Functional capability, project management, material and environmental functionality
DR	Calculation and data requirements
UX	User interaction and localization requirements
EX	Reports and export requirements

Each numbered requirement includes a statement, priority, verifiable acceptance criteria and current prototype status. **Must** denotes essential functionality approved within the product scope by Knauf Commercial / the designated product owner, with existing reference behavior used to make that functionality testable. This specification does not introduce speculative features as Should or Could requirements. Unapproved extensions and unresolved policy details appear in section 12 without requirement priorities.

Prototype status has the following meaning:

• **Implemented:** the relevant behavior is present in the reviewed code; this is not a claim of production acceptance or independent engineering verification.

• **Partial:** some behavior is present, but an identified gap prevents full compliance with the requirement.

• **Gap:** the approved target behavior is not implemented.

• **Implemented with input-validation limitations:** the calculation logic is implemented; numeric input validation still needs improvement.

1.2 Calculation and metadata semantics

A **current calculation** is the calculation being entered or edited. A **saved calculation** is an entry explicitly added to or updated within a project. A **saved material snapshot** is the material result stored by that Add/Update action. Saving a project file preserves the project; it is not equivalent to updating its calculations.

The latest entered project name, author, company and description must be accepted as part of a report-download action without a separate field-application step. This does not silently update saved engineering results or saved material snapshots. Individual exports concern the current valid calculation; project exports concern the explicitly saved calculations. This distinction is approved; OQ-10 concerns only its detailed presentation to users.

2. Scope and Intended Users

Fireboard is a digital calculation tool for dimensioning fire protection of steel structures using Knauf Fireboard. Target users include architects, consulting engineers, contractors, technical consultants, estimators and building-material distributors. All users follow the same basic calculation workflow.

The solution supports:

- Fire-protection dimensioning for standard and custom steel profiles.
- Encasement methods, fire resistance periods and design steel temperatures.
- Calculation of required Fireboard thickness and indicative material quantities where supported.
- CO_2 information and product documentation.
- Projects containing multiple calculations, including saving, editing and reopening projects.
- PDF, Excel, ZIP and separate editable project-file exports.
- Danish and English localization.

The design steel temperature is determined by the responsible designer or consultant and entered as a calculation input. Custom-profile material limitations are defined in section 4.2.

3. User Workflow

Step	User activity	Functional output
Profile	Select a standard category and size, or define a custom profile and its Ap/V input method.	Profile identity and basis for the profile factor.
Cladding	Select a cladding method and the number of protected sides.	A supported protection arrangement and, where applicable, completed geometry-based Ap/V.
Fire Requirements	Select R30, R60, R90 or R120 and enter the externally determined design steel temperature.	Inputs for the appropriate thickness data.
Result	Review Ap/V and thickness, project information, and applicable material results; add/update a calculation or export.	A valid calculation result and the permitted project/export actions.

UX-001	
Requirement	Provide the four-step Profile, Cladding, Fire Requirements and Result workflow, with understandable navigation and input validation.
Priority	Must
Acceptance criteria	Users can move between the four steps without losing applicable inputs. Opening Result with incomplete or unsupported inputs gives an actionable explanation and does not present an invalid thickness as a valid result.
Prototype status	Implemented workflow and checks; localization of some validation messages is partial.

Returning to an earlier step permits review and editing. The production application need not reproduce the prototype's exact navigation widgets or page arrangement. Any changed input must be reflected in the current calculation before that result is presented or exported as current.

4. Functional Requirements

4.1 Profiles and fire dimensioning

FR-001	
Requirement	Allow selection of standard steel profiles from the supplied category-specific catalogue.
Priority	Must
Acceptance criteria	The catalogue exposes H profiles (HEA/HEB/HEM), I profiles (IPE/INP), U profiles (UNP), hot- and cold-rolled square and rectangular hollow sections, and medium and heavy circular hollow sections. The selected category and size remain distinguishable throughout calculation and reporting.
Prototype status	Implemented. Catalogue content is data-driven; the reviewed baseline has 3,050 distinct category/profile/method/sides configurations.

FR-002	
Requirement	Support named custom profiles with directly entered Ap/V.
Priority	Must
Acceptance criteria	A user can name a custom profile and enter a positive Ap/V in m^2/m^3. Round the value upward to the nearest whole number before thickness lookup; a result requires that rounded value to be at most 380 and a valid thickness to exist. Saved custom names and values can be restored.
Prototype status	Partial: entry and restoration exist, but the prototype truncates fractional direct Ap/V instead of applying the approved upward rounding.

FR-003	
Requirement	Support custom profiles whose Ap/V is determined from the reference geometry inputs.
Priority	Must
Acceptance criteria	Users can enter a and b in mm and steel area A in mm^2, select protected sides, obtain Ap/V and review guidance for the input convention. Round positive calculated Ap/V upward to the nearest whole number before thickness lookup and apply the maximum of 380. Invalid or incomplete geometry cannot produce a valid result.
Prototype status	Partial: geometry and Danish/English guidance exist, but the prototype rounds Ap/V to the nearest integer rather than always upward. It is a prescribed geometry convention, not arbitrary shape modelling.

FR-004	
Requirement	Support clamp cladding and beam/PHL profile cladding with the permitted protected-side configurations.
Priority	Must
Acceptance criteria	The selected method and sides are included in calculation matching. Unsupported combinations are rejected. Under the current data, circular standard profiles support clamp cladding on four sides; other categories support only combinations actually present in the data.
Prototype status	Implemented lookup validation. Some unsupported choices remain visible until validation.

FR-005	
Requirement	Support fire resistance durations R30, R60, R90 and R120.
Priority	Must
Acceptance criteria	Each duration selects the corresponding approved thickness data. The selected duration appears in results, saved calculations and calculation reports.
Prototype status	Implemented.

FR-006	
Requirement	Accept design steel temperature as an external engineering input and explain its source and responsibility.
Priority	Must
Acceptance criteria	Users enter the temperature supplied by the structural designer or responsible consultant. The application does not calculate it or imply it has verified structural adequacy. Values must be supported by the calculation data; the current reference accepts whole degrees from 350 to 750 °C inclusive.
Prototype status	Partial: input and range validation exist; the complete approved responsibility guidance is not established in the UI.

FR-007	
Requirement	Present the calculated Ap/V and required total Fireboard thickness for valid inputs.
Priority	Must
Acceptance criteria	A supported case displays Ap/V in m^2/m^3 and thickness in mm together with the selected engineering inputs. Unsupported or incomplete cases do not receive a valid thickness result. A changed input cannot be presented alongside an unrelated previous result.
Prototype status	Implemented dimensioning and result display; some error messages remain unlocalized.

4.2 Custom-profile functional boundary

Custom profiles support thickness calculations, calculation reports and project persistence. The prototype does not calculate their material consumption or material CO_2, including for geometry-based custom profiles. This draft does not approve an extension to custom-profile quantities. Exports must identify unavailable quantities rather than imply that the requirement is zero; see FR-020 and EX-004.

5. Project Management Requirements

FR-008	
Requirement	Maintain shared project metadata comprising project name, prepared by, company and multiline description.
Priority	Must
Acceptance criteria	Metadata can be edited, saved and reopened. Generated calculation reports use the project information applicable to the export action. No user-entered report date overrides the generation date.
Prototype status	Partial: fields and persistence exist; immediate-download synchronization has a confirmed gap. The redundant date field has been removed.

FR-009	
Requirement	Allow a project to contain multiple independent calculations, with one calculation able to document multiple profiles under identical engineering conditions.
Priority	Must
Acceptance criteria	Adding a new valid calculation preserves existing entries and their stored material results. Standard and custom profiles can coexist. Ten HEB100 profiles of 6 m each, with identical category, size, fire duration, design temperature, cladding method and protected sides, can use one calculation and a combined material-estimation length of 60 m. No separate profile-count input or duplicate calculations are required for this case.
Prototype status	Implemented project entries and length-based estimates support the approved approach. Exact repetition of the most recently added calculation is suppressed by the prototype; no new duplicate-calculation feature is required.

FR-010	
Requirement	Restore the selected saved calculation for review and editing.
Priority	Must
Acceptance criteria	Selecting calculation A after B restores A's category, profile/name, method, protected sides, fire inputs and material settings. Custom profiles restore their Ap/V method and relevant direct or geometry values. Other saved entries remain intact.
Prototype status	Implemented, including corrected custom-profile restoration.

FR-011	
Requirement	Provide an explicit Update Calculation action that replaces the selected saved calculation and its applicable material snapshot.
Priority	Must
Acceptance criteria	Updating an existing entry changes that entry without increasing the project count. Its saved result and material quantities correspond to the reviewed valid inputs. Other calculations are unchanged.
Prototype status	Implemented with regression coverage.

FR-012	
Requirement	Allow deletion of a selected saved calculation after confirmation.
Priority	Must
Acceptance criteria	Confirmation identifies the intended deletion. Cancel leaves the project unchanged; confirm removes only that entry and removes its contribution from combined material quantities.
Prototype status	Partial: deletion and rebuilding totals exist; confirmation is a gap.

FR-013	
Requirement	Provide New Calculation as an independent current calculation without modifying saved project calculations.
Priority	Must
Acceptance criteria	After New Calculation, editing and adding the new calculation does not overwrite an existing entry. Saved calculations and shared project metadata remain intact.
Prototype status	Implemented independence. Some custom input values can remain visible; this prototype behavior was accepted and is not an unresolved defect. A fully blank draft is not imposed by this requirement.

FR-014	
Requirement	Provide New Project to clear the current project after confirmation.
Priority	Must
Acceptance criteria	Cancel preserves current project information and calculations. Confirm clears project calculations and metadata and resets the current workflow to its initial state. Previously downloaded local files are unaffected.
Prototype status	Partial: clearing/resetting exists, but the prototype acts without confirmation.

FR-015	
Requirement	Allow projects created by the production solution to be saved locally and reopened, with understandable handling of invalid or incompatible files.
Priority	Must
Acceptance criteria	A project created by the production solution round-trips project metadata, multiple calculations, material settings and stored snapshots. An invalid file produces a clear explanation. Saving and reopening preserve saved results without recalculation.
Prototype status	Implemented in the prototype: local save/open and structural validation. The production file round trip requires production acceptance.

FR-016	
Requirement	Preserve previously saved results on opening, without automatic recalculation against newer data.
Priority	Must
Acceptance criteria	Opening a project saved by the production solution retains its stored Ap/V, thickness and material quantities even after the calculation dataset changes. Users must select, review and explicitly update an entry to apply current data. Unmodified entries remain unchanged.
Prototype status	Implemented preservation. Dataset revision identification and presentation of saved results are limited.

FR-017	
Requirement	Preserve each calculation's applicable material settings while navigating and switching calculations.
Priority	Must
Acceptance criteria	Profile length, waste and valid selected build-up are restored for the selected calculation. Navigating away and back does not silently discard these settings or copy another calculation's settings.
Prototype status	Implemented. Retained settings and saved quantities can differ until Update Calculation.

Fireboard has not yet been commercially launched. Project persistence concerns projects created by the production solution. Saved engineering and material results must remain unchanged when projects are reopened, saved or exported, including after a dataset update; applying current data requires explicit review and Update Calculation.

6. Material and Environmental Requirements

FR-018	
Requirement	Offer the valid material build-up variants for the required total Fireboard thickness.
Priority	Must
Acceptance criteria	Only variants supported by the supplied layer rules are selectable. Changing a variant updates applicable products and quantities without changing the required total thickness. A previously selected variant is retained only if still valid.
Prototype status	Implemented.

FR-019	
Requirement	Support decimal profile length in metres and decimal waste percentage as material-estimate inputs, defaulting to 6.0 m and 10%.
Priority	Must
Acceptance criteria	Length must be greater than 0 m; waste must be at least 0%, with no separate upper waste limit. For multiple profiles with identical engineering conditions, use their combined total length. Both inputs and units remain visible and persist with the calculation. Invalid input produces a clear validation message and never silently becomes zero.
Prototype status	Implemented with input-validation limitations. Defaults and quantity formulas are implemented; range checks and rejection of invalid numeric text require improvement.

FR-020	
Requirement	Refresh the current material result when length, waste or build-up changes, where material calculation is supported.
Priority	Must
Acceptance criteria	After changing a valid input, the current table, applicable CO_2 and individual material export agree with that input. Invalid length or waste is explained rather than presented as a valid zero-quantity estimate. For custom profiles without material support, the application explicitly reports that quantities and material CO_2 are unavailable.
Prototype status	Implemented with input-validation limitations. Refresh and custom-profile exclusion are implemented. Project workbook notes can still show numeric zero placeholders; unavailable quantities must be identified explicitly.

FR-021	
Requirement	Update saved project material snapshots only through explicit Add/Update Calculation actions.
Priority	Must
Acceptance criteria	For a saved 6 m calculation, editing the current length to 12 m refreshes the current estimate but leaves the saved 6 m material snapshot unchanged until Update Calculation. Saving/reopening the project alone does not substitute the 12 m quantities.
Prototype status	Implemented. The distinction needs clearer user communication.

FR-022	
Requirement	Provide combined material quantities from saved project material snapshots and preserve a per-calculation breakdown.
Priority	Must
Acceptance criteria	Combined totals equal the sum of included saved quantities for each material and unit. Deleting/updating an entry changes only its contribution. An unsaved current draft is not silently included.
Prototype status	Implemented totals and breakdown exports, with identity/localization limitations described in section 11.

FR-023	
Requirement	Distinguish actual materials independently of translated descriptions and intentionally blank displayed article numbers.
Priority	Must
Acceptance criteria	Distinct 50 mm and 64 mm clamps remain separate even when both display blank article numbers. Repeated identical clamps aggregate. Switching language cannot change material identity or split/merge quantities incorrectly.
Prototype status	Partial: corrected clamp catalogue and current-list aggregation are covered by tests; combined exports still fall back to displayed descriptions when article numbers are blank.

FR-024	
Requirement	Clearly label material lists as indicative material consumption.
Priority	Must
Acceptance criteria	Material lists use “Vejledende materialeforbrug” in Danish and “Indicative material consumption” in English. Quantities are not presented as optimized board counts, stock lengths or package orders.
Prototype status	Implemented. The prototype already identifies material consumption as indicative.

FR-025	
Requirement	Present product identification, quantities, CO_2 information and available supporting-document links for applicable materials.
Priority	Must
Acceptance criteria	Material results identify the product, manufacturer, available article/DB numbers, unit, per-length rate and total quantity. Applicable numeric CO_2 uses kg CO_2e; available EPD and datasheet links are accessible, and missing links are not fabricated.
Prototype status	Implemented current tables and links, subject to localization and catalogue-quality limitations.

FR-026	
Requirement	Calculate material CO_2 using factors from Knauf's existing product database, based on product EPDs for lifecycle modules A1–A3 and C3–C4.
Priority	Must
Acceptance criteria	CO_2 uses calculated material quantities including applicable waste allowances and the product-specific bases in DR-007–DR-008. Screws, staples/clamps, joint tape and other specifically excluded components remain in material lists but make no numeric CO_2 contribution under the applied BR18 methodology. Apply the defined product mapping; do not assume all minor products are excluded.
Prototype status	Implemented. Calculation logic and defined exclusions are covered by automated tests, including repeated-layer aggregation and excluded products.

The current layer data provides the following reference build-ups. These are data content, not a requirement to hardcode the choices in the future application.

Total thickness	Supported reference build-ups
15 mm	15 mm
20 mm	20 mm
25 mm	25 mm
30 mm	30 mm; 15 + 15 mm
35 mm	20 + 15 mm
40 mm	20 + 20 mm
45 mm	25 + 20 mm; 30 + 15 mm
50 mm	25 + 25 mm; 30 + 20 mm

The catalogue at the reviewed baseline contains 23 unique article numbers. Article 90000005 is the minimum 50 × 9 × 1.5 mm steel clamp; 90000006 is the minimum 64 × 9 × 1.5 mm clamp. The former is selected for applicable 20 mm board constructions. These products are useful acceptance fixtures for identity and aggregation.

7. Reports and Export Requirements

EX-001	
Requirement	Export an individual fire-protection dimensioning PDF for the current valid calculation, including custom profiles.
Priority	Must
Acceptance criteria	The report identifies project metadata, profile category/name, cladding method, protected sides, fire duration, design temperature, Ap/V and required Fireboard thickness. It can be generated without first adding the calculation to the project. It does not include material tables or CO_2 schedules.
Prototype status	Implemented approved report content; latest-entered metadata at immediate download is a gap.

EX-002	
Requirement	Export a combined fire-protection dimensioning PDF for saved project calculations.
Priority	Must
Acceptance criteria	The report includes each saved calculation with its stored engineering result, shared project metadata and clear page numbering. It excludes an unsaved draft, material tables and CO_2 schedules, and does not automatically recalculate historical results.
Prototype status	Implemented one page per saved calculation and approved content boundaries; metadata synchronization is partial.

EX-003	
Requirement	Export the current supported indicative material list to Excel.
Priority	Must
Acceptance criteria	The workbook includes product identification, manufacturer, description, per-linear-metre consumption, unit and total quantity matching the current valid material result. Unsupported custom-profile quantities are not invented. CO_2, EPD and datasheet-link columns are excluded.
Prototype status	Implemented approved content boundaries; localization remains partial.

EX-004	
Requirement	Export combined indicative project material quantities and a per-calculation material breakdown to Excel.
Priority	Must
Acceptance criteria	Preserve both material-list options. Both views reconcile to saved material snapshots. Identical products combine correctly; distinct products remain distinct. Custom-profile or missing-snapshot quantities are identified as unavailable, not calculated zero. CO_2, EPD and datasheet-link columns are excluded.
Prototype status	Partial: both exports and approved content boundaries exist; description-based grouping, untranslated notes and zero placeholders need correction. The prototype's summed per-metre-rate column is not a meaningful project-wide consumption rate.

EX-005	
Requirement	Export a project documentation ZIP containing the calculation report, applicable project material workbooks and available supporting documentation.
Priority	Must
Acceptance criteria	The package contains the combined PDF, combined and per-calculation material workbooks where applicable, Fireboard design/installation sections, and applicable product EPD/datasheet documents. It uses the same saved project basis as the corresponding standalone exports. The editable .fireboard file is not automatically included.
Prototype status	Implemented approved package structure; latest metadata and localization are partial.

EX-006	
Requirement	Handle unavailable external package documents explicitly without misrepresenting package completeness.
Priority	Must
Acceptance criteria	A failed or empty external-document download does not silently appear as a successful inclusion. The generated package identifies omitted documents in the selected application language. Repeated product-document references do not create unnecessary duplicates of the same document type.
Prototype status	Implemented skipped-file README and URL deduplication; some descriptive filenames and underlying error text are not fully localized.

EX-007	
Requirement	Export and reopen editable project data separately from PDF, Excel and ZIP documentation.
Priority	Must
Acceptance criteria	The project file preserves metadata, current workflow information, saved calculations, material settings and stored snapshots required for continued review. Saving the file does not recalculate results or imply that a ZIP contains editable project data.
Prototype status	Implemented separate .fireboard export in the prototype. Production files must satisfy FR-015–FR-016.

EX-008	
Requirement	Display the PDF generation date in Europe/Copenhagen using DD-MM-YYYY, consistently across the generated report.
Priority	Must
Acceptance criteria	Test generation around UTC midnight and both daylight-saving transitions. All pages of a combined report have the same captured date. Downloading an unchanged previously generated/cached file later retains its original generation date.
Prototype status	Implemented and covered by automated tests. Live deployment timezone-data availability has not been directly verified.

EX-009	
Requirement	Wrap project descriptions within the approved current PDF layout, with at most two lines and visible ellipsis for overflow.
Priority	Must
Acceptance criteria	Short text retains its established position. Longer text wraps by rendered width; explicit line breaks are respected where possible; long words/URLs do not exceed the cell. Overflow ends with a fitted ellipsis. Danish characters and literal <, > and & are preserved. The stored full description is unchanged. Applies to both languages and PDF types.
Prototype status	Implemented shared rendering and coordinate-aware tests against both templates.

EX-010	
Requirement	Ensure every generated export uses a coherent calculation/material snapshot and the project metadata accepted for that export action.
Priority	Must
Acceptance criteria	Individual outputs reflect the current valid calculation; project outputs reflect explicitly saved calculation/material results. Latest entered metadata is included without a separate Apply action. A prior export must not be served as current after relevant metadata changes; unchanged historical files retain their original bytes/date.
Prototype status	Partial: calculation/snapshot boundaries and content-based caches exist, but the first download can contain stale metadata.

7.1 Existing report design and content boundaries

The existing Danish and English PDF templates establish the approved description area and surrounding fields. Preserve the established appearance for short descriptions, fit wrapped text within that area and prevent overlap with neighbouring content. The reference templates and layout tests support visual acceptance without prescribing a rendering library or coordinate-based implementation.

Knauf Commercial / the designated product owner has approved preserving the existing export separation and content as functional scope. Individual and combined PDFs document fire-protection dimensioning, while individual, per-calculation and combined Excel material lists contain indicative material quantities. The ZIP combines calculation documentation, material lists and available supporting product documentation; the separate .fireboard file preserves editable project data.

Calculation PDFs do not include material tables or CO_2 schedules; the prototype also omits length/waste controls and custom geometry dimensions. Material Excel exports exclude CO_2, EPD and datasheet-link columns. The ZIP does not automatically include an editable .fireboard file and currently has no separate individual-calculation PDFs. No additional export types or content extensions are authorized by this specification.

The combined workbook currently sums per-linear-metre rates as well as total quantities. This is a prototype behavior: the summed rate must not be interpreted as a meaningful project-wide consumption rate. FR-022 requires correct combined total quantities. This specification introduces no alternative rate-aggregation formula.

8. Localization and User Experience

UX-002	
Requirement	Support Danish and English for all application-controlled user-facing content.
Priority	Must
Acceptance criteria	Switching language localizes workflow labels, guidance, validation, errors, warnings, confirmations, material notes, report labels and application-authored package explanations. Native framework/system messages may remain English where outside practical application control.
Prototype status	Partial: principal labels and PDF templates are bilingual; several validation and export messages remain Danish.

UX-003	
Requirement	Obtain Danish and English product descriptions and product units from Knauf EPIM through a PRV export.
Priority	Must
Acceptance criteria	Descriptions and units are maintained centrally in EPIM. The PRV file is uploaded with the other Fireboard data files whenever the dataset is updated, using a file-based workflow similar to SystemFinder. Products resolve through stable product identifiers. Switching language selects the corresponding description and unit without changing product identity, material quantities or calculation results. Missing mandatory descriptions or units prevent publication under DR-014.
Prototype status	Not implemented. The prototype uses a local material catalogue; the EPIM/PRV update workflow is new.

UX-004	
Requirement	Present all CO_2 results and material quantities with exactly two decimal places, using Danish decimal commas and English decimal points.
Priority	Must
Acceptance criteria	Accept both comma and period as decimal separators for numeric user input in both languages. Show the same quantity as 12,50 in Danish and 12.50 in English, including trailing zeros. Apply this presentation to applicable screens and exports. Retain sufficient precision internally; compute totals from unrounded values and round only for display so display rounding introduces no errors into totals. Language changes preserve numeric values. Ap/V follows DR-002 separately.
Prototype status	Partial: two-decimal material formatting exists but uses commas in both languages; numeric input and total precision require improvement.

UX-005	
Requirement	Localize Excel worksheet names and application-controlled descriptive report/document filenames.
Priority	Must
Acceptance criteria	English exports use English worksheet/descriptive filenames and Danish exports use Danish equivalents. Application-authored custom-profile notes follow the selected language. User-entered portions of filenames are not automatically translated.
Prototype status	Partial: some titles and names are localized; worksheet names and several product-document filenames/notes remain Danish.

UX-006	
Requirement	Preserve technical identifiers and user-authored names across language changes and project reopening.
Priority	Must
Acceptance criteria	Article numbers, material codes, internal keys and the SYSTEM heading remain unchanged. User-entered or previously saved profile names and project text are not automatically translated. Display-label changes cannot alter engineering matching.
Prototype status	Partial: names and main technical identifiers are retained; display-language material descriptions are still involved in some grouping/resolution paths.

UX-007	
Requirement	Accept the latest entered project metadata as part of the download action, without Ctrl+Enter or an unnecessary per-field Save/Apply step.
Priority	Must
Acceptance criteria	With focus still in a changed multiline description, the user clicks Download Calculation directly; the first downloaded PDF contains the latest entered value, subject only to approved visible truncation. Repeat for combined PDFs and ZIP-contained reports, including existing cached downloads and clearing a field. No additional preparation/apply action is introduced solely to accommodate prototype framework behavior.
Prototype status	Gap: immediate downloads can use previously generated bytes. Sidebar project exports also precede the durable metadata update in the prototype.

UX-008	
Requirement	Require confirmation before deleting a calculation or clearing the project with New Project.
Priority	Must
Acceptance criteria	The confirmation explains the affected calculation or project. Cancel has no destructive effect; confirming performs only the stated action.
Prototype status	Gap: both actions currently execute immediately. This is an approved target requirement, not an unresolved product decision.

UX-009	
Requirement	Make the difference between current edits, saved calculations and saved material snapshots clear and predictable.
Priority	Must
Acceptance criteria	Users can identify which calculation is being edited and whether project quantities require Update Calculation. An update does not unexpectedly add an entry. New Calculation does not silently modify an existing saved entry. Report-generation metadata acceptance is not confused with updating saved material quantities.
Prototype status	Partial: add/update modes and independent calculations exist; snapshot freshness and export timing are insufficiently explicit.

UX-010	
Requirement	Explain invalid, incomplete or unsupported inputs in the selected language without presenting diagnostics as normal user guidance.
Priority	Must
Acceptance criteria	Missing engineering inputs, an unsupported configuration, upward-rounded Ap/V above 380 or an unavailable thickness produce an actionable localized message and no valid result for that case. Length at or below zero, negative waste and invalid numeric input produce clear validation messages; invalid input never silently becomes zero. Valid decimal length/waste and waste above 100% are not rejected merely for being fractional or above 100%. Application-controlled debug details are not the user's only explanation.
Prototype status	Partial: checks exist, but Ap/V rounding and length/waste validation do not meet the approved rules; some messages are hardcoded Danish and one unsupported lookup path displays debug details.

External EPDs, datasheets and technical documents are not translated automatically. Their source language and content remain those of the supplier. This is distinct from localizing the application's link labels, explanatory notes and descriptive download names.

Production requirements must not inherit a requirement to press a particular framework shortcut, add an Apply button, or accept a second download attempt. UX-007 remains a target outcome for Knauf Digital to implement in its chosen technology. The prototype's synchronization limitation does not change that approved outcome.

9. Calculation and Data Requirements

DR-001	
Requirement	Match standard-profile Ap/V and material-consumption data using category, profile, cladding method and protected sides together.
Priority	Must
Acceptance criteria	All 3,050 reference configurations resolve to their own records independently of source row order. Cross-category name collisions never select another category. Missing configurations remain unsupported rather than falling back to a category-free match.
Prototype status	Implemented correction and exhaustive reference tests; no duplicate four-field keys at the baseline.

DR-002	
Requirement	Round every positive Ap/V upward to the nearest whole number before thickness lookup and enforce a maximum of 380 m^2/m^3 for every standard and custom calculation and supported fire duration.
Priority	Must
Acceptance criteria	Exact integers remain unchanged. Verify 125.0 → 125, 125.1 → 126, 125.7 → 126, 379.9 → 380, 380.0 → 380 and 380.1 → 381 → rejected. Test direct and geometry values across R30/R60/R90/R120 and both languages. Values 379 and 380 require a valid table cell; 381, 400 and 401 are rejected. A table entry above 380 cannot override the maximum.
Prototype status	Partial: the integer maximum is implemented, but direct truncation and nearest-integer geometry rounding differ from the approved upward-rounding rule. Standard integer catalogue values are unchanged by upward rounding.

DR-003	
Requirement	Determine thickness only from the applicable approved fire-duration data and supported upward-rounded Ap/V/temperature coordinates.
Priority	Must
Acceptance criteria	Results match the reference fixtures in section 10. Missing coordinates or blank cells are rejected even when rounded Ap/V is at or below 380. Apply DR-002 before lookup; rounding does not authorize interpolation, extrapolation or substitution for missing thickness data.
Prototype status	Partial: exact table lookup exists, but the prototype's preceding Ap/V normalization differs from DR-002.

DR-004	
Requirement	Preserve the physical units and reference input meaning of custom-profile Ap/V, applying approved upward rounding to the resulting value.
Priority	Must
Acceptance criteria	Direct input is in m^2/m^3. Geometry inputs are a/b in mm and A in mm^2. Invalid non-positive or incomplete geometry yields no result. Reference geometry examples reproduce the perimeter-to-area calculation described below, then round the positive result upward under DR-002 without prior nearest-integer rounding or truncation.
Prototype status	Partial: prescribed geometry is implemented; its final nearest-integer rounding must become upward rounding in production. Engineering confirmation of the reference geometry guidance is tracked under OQ-04.

DR-005	
Requirement	Resolve material build-ups and associated products using the applicable layer and component rules.
Priority	Must
Acceptance criteria	Each offered variant totals the required thickness. Screws/clamps are resolved by the appropriate board layer/thickness; beam and angle materials follow the relevant profile/side rules. Unsupported combinations do not create arbitrary products.
Prototype status	Implemented using layer, fastener, angle and beam data.

DR-006	
Requirement	Preserve the reference quantity basis, including combined profile length, waste and repeated layers.
Priority	Must
Acceptance criteria	For decimal length $L > 0$ and decimal waste $w \geq 0$, an aggregated per-metre rate r produces $r \times L \times (1 + w/100)$ in the applicable unit, without a separate upper waste limit. L can be the combined length of identical profiles under FR-009; ten 6 m members use L = 60 m without a separate count multiplier. Repeated layers contribute their full material rates. Finishing products retain their reference area basis rather than being multiplied indiscriminately by layer count. Invalid numeric input produces a clear validation error and never silently becomes zero. Accept both decimal separators under UX-004; retain full calculation precision until display.
Prototype status	Implemented with input-validation limitations.

DR-007	
Requirement	Apply area-based Fireboard CO_2 factors to the full aggregated board area including waste.
Priority	Must
Acceptance criteria	For repeated identical Fireboard layers, reported area and CO_2 both include every layer. For an area factor f and aggregated area Q including waste, CO_2 equals $Q \times f$ before display rounding. Aggregating identical rows cannot discard their emissions.
Prototype status	Implemented correction with repeated-layer regression tests.

DR-008	
Requirement	Apply product-specific environmental factors from Knauf's product database, based on product EPDs covering A1–A3 and C3–C4, with the defined CO_2 bases and exclusions.
Priority	Must
Acceptance criteria	Linear products use the appropriate total linear quantity; area-based products use their prescribed area basis; filler retains its reference-area treatment. Include applicable waste. Excluded products remain in material lists without numeric CO_2 contributions. Compute totals from included unrounded values and display CO_2 with exactly two decimals under UX-004.
Prototype status	Product-specific calculation logic and exclusions are implemented and tested. Presentation and total precision have the limitations recorded under UX-004.

DR-009	
Requirement	Aggregate materials by stable actual identity and compatible units, independently of presentation language.
Priority	Must
Acceptance criteria	Merging calculations with the same product in Danish and English preserves the summed quantity and applicable CO_2. The two clamp articles remain distinct despite blank display numbers. Reordering data or translating labels does not change totals. Product descriptions and units come from EPIM/PRV; stable product identifiers govern aggregation, independently of translated descriptions.
Prototype status	Partial: current-list deduplication is covered; description fallback in combined exports remains a gap. EPIM/PRV handling is not implemented.

DR-010	
Requirement	Preserve the logical relationships between profile data, thickness data, component rules and the product catalogue.
Priority	Must
Acceptance criteria	Every supported standard lookup has an unambiguous four-field identity. Products referenced by component rules resolve to the intended catalogue item. Data inconsistencies are not resolved by selecting an arbitrary conflicting record or inventing an engineering result. Product relationships use stable identifiers; descriptions and units come from EPIM/PRV. Calculation logic is independent of display-language descriptions. Validate these relationships before publication under DR-014.
Prototype status	Current dataset has 3,050 unique configuration keys, 23 unique catalogue articles and corrected clamp references. Comprehensive production data-release validation is not established.

DR-011	
Requirement	Apply newer calculation data to a saved entry only after explicit user review and Update Calculation.
Priority	Must
Acceptance criteria	A saved production project opens with its original results. Reviewing and updating one selected calculation uses current applicable data and replaces that entry's result/snapshot; other saved entries remain unchanged. Merely exporting or saving the project file does not recalculate it.
Prototype status	Implemented preservation/update behavior. Saved files do not contain a calculation-data revision identifier; introducing one is not mandated by this specification.

9.1 Engineering reference conventions

The geometry reference uses exposed perimeter P = a + 2b for one or three protected sides, P = a + b for two sides, and P = 2a + 2b for four sides. The unrounded Ap/V is P × 1,000 / A. These conventions require the matching input guidance; they must not be represented as a universal model for every steel shape. Confirmation of that guidance as part of the engineering reference data remains under OQ-04.

Approved production rule: round every positive Ap/V upward to the nearest whole number, leaving exact integers unchanged. For geometry, apply this rule to the unrounded calculated value; for direct entry, apply it to the entered value. Validate the resulting integer against the maximum of 380 before looking up thickness. This rule applies across R30, R60, R90 and R120 and does not change category-specific selection or geometry formulas. Thus 379.9 becomes 380 and may receive a thickness if a valid cell exists, while 380.1 becomes 381 and is rejected.

Prototype difference: the current code truncates fractional direct Ap/V and rounds geometry-derived Ap/V to the nearest integer before checking the maximum. It therefore does not yet satisfy the approved production rounding rule. This documentation update does not change that code.

The current temperature reference is 350–750 °C in whole-degree values. The Fireboard thickness result is a total thickness in mm; layer selection is a subsequent material-build-up operation. Results outside the supported data remain unavailable even if they fall within a general input range.

For values above the maximum, the approved messages are:

• Danish: “Profilforholdet (Ap/V) overstiger 380 m²/m³, vælg et andet eller større profil”

• English: “The profile factor (Ap/V) exceeds 380 m²/m³. Select another or larger steel profile.”

9.2 Reference data inventory

These files identify the current reference content. CSV storage and these filenames are not prescribed as the production data architecture.

Reference data	Purpose and relationship
apv.csv	Standard profile categories/names, cladding and sides, Ap/V, and rates for Fireboard area, beam/PHL, angles, screws and clamps.
fireboard_30.csv, fireboard_60.csv, fireboard_90.csv, fireboard_120.csv	Thickness values by design temperature and Ap/V for each duration. Underlying columns beyond 380 do not authorize their use.
layer_logic.csv	Valid layer thicknesses/order and build-up variants for a required total thickness.
screw_clamp_logic.csv	Layer/board-thickness rules referring to screw and clamp catalogue products.
angle_profile_logic.csv	Angle-profile consumption associated with protected sides.
beam_profile_logic.csv	Profile-related beam/PHL product selection.
materials.csv	Prototype local catalogue: product identity, manufacturer, Danish/English descriptions, CO_2 factors/types and document links; resolves products selected by component rules. Production descriptions and units are centrally maintained in EPIM and supplied through PRV. Environmental factors originate from Knauf’s product database and product EPDs.

Knauf Commercial – Local Data Owner maintains, uploads and publishes calculation and product datasets. EPIM is the central source for product descriptions and units; PRV is uploaded with the related Fireboard files for each update. Environmental factors originate from Knauf’s product database and product EPDs. Knauf Digital provides the application and technical operation. Sections 9.3 and 12 define the responsibilities and publication controls.

9.3 Data Administration and Self-Service Upload Portal

The production update workflow is file-based, similar to SystemFinder: an authorized Local Data Owner uploads the EPIM PRV export together with the related Fireboard data files, reviews automatic validation and publishes a valid dataset. No live EPIM API integration is required. These five requirements extend the original 57 requirements to 62.

DR-012	
Requirement	Provide a self-service administration portal for an authorized Local Data Owner.
Priority	Must
Acceptance criteria	The Local Data Owner can upload, validate and publish updated Fireboard datasets independently. No manual intervention or per-update approval from Knauf Digital is required.
Prototype status	Not implemented.

DR-013	
Requirement	Support uploading the EPIM PRV export together with the relevant Fireboard data files as one dataset update.
Priority	Must
Acceptance criteria	The update includes PRV product descriptions and units, profile data, fire-protection thickness tables, material and component rules, and relevant environmental data. All related files are validated together before publication.
Prototype status	Not implemented.

DR-014	
Requirement	Automatically validate datasets before publication and reject invalid datasets.
Priority	Must
Acceptance criteria	Validate file structure and mandatory columns; required Danish and English product descriptions and product units; stable product identifiers; relationships between products, material rules and calculation datasets; and missing or inconsistent mandatory information. Missing mandatory descriptions or units prevent publication. The actionable validation report identifies affected files, products and fields and explains the required correction.
Prototype status	Not implemented.

DR-015	
Requirement	Publish valid datasets safely and protect the active production dataset.
Priority	Must
Acceptance criteria	A failed upload or validation does not modify the active dataset. The latest successfully published dataset remains operational until a new valid dataset is published. Publication never exposes a partially updated dataset. Saved project results remain preserved under FR-016 and DR-011.
Prototype status	Not implemented.

DR-016	
Requirement	Enforce authorized data-administrator access and the agreed operational responsibilities.
Priority	Must
Acceptance criteria	Knauf Commercial – Local Data Owner maintains, uploads and publishes calculation and product datasets. Knauf Digital owns the application, technical architecture, hosting, security, maintenance, troubleshooting and operational availability. Appropriate access controls restrict administration to authorized users. EPIM remains the central source for product descriptions and units.
Prototype status	Not implemented.

10. Acceptance Criteria

The criteria in each requirement row are normative for this specification. The following integrated scenarios support acceptance across calculation, project, export and language boundaries. Prototype test results are supporting evidence, not a substitute for production acceptance in the technology selected by Knauf Digital.

10.1 Calculation reference fixtures

The approved upward-rounding rule adds the following production acceptance cases. Exercise each value as direct input and as an unrounded geometry result, for all four fire durations and both languages. Permitted values still require a valid thickness at the rounded table coordinate.

Positive Ap/V before rounding	Required lookup Ap/V	Limit outcome
125.0	125	Permitted if the table cell is valid
125.1	126	Permitted if the table cell is valid
125.7	126	Permitted if the table cell is valid
379.9	380	Permitted if the table cell is valid
380.0	380	Permitted if the table cell is valid
380.1	381	Rejected before thickness lookup

The following existing integer reference fixtures remain applicable; upward rounding leaves their Ap/V unchanged.

Profile and arrangement	Required outcome
Heavy circular 100 mm, clamp, four sides	Ap/V 130 m^2/m^3; R60 / 350 °C gives 25 mm.
Medium circular 100 mm, clamp, four sides	Ap/V 154 m^2/m^3; it must not inherit the heavy category's value.
Cold-rolled square 100 × 100 × 4 mm, beam/PHL, three sides	Ap/V 242 m^2/m^3; R60 / 750 °C gives 20 mm.
Cold-rolled rectangular 100 × 60 × 5 mm, beam/PHL, three sides	R90 / 550 °C gives 35 mm.
Cold-rolled rectangular 180 × 100 × 5 mm, beam/PHL, four sides	Ap/V 258 m^2/m^3; R120 / 750 °C gives 40 mm.
HEB100 and IPE120, clamp, three sides	Ap/V remains 116 and 231 m^2/m^3 respectively.
Ap/V 379 and 380, temperature 750 °C	Valid reference thicknesses: R30 15 mm; R60 30 mm; R90 30 mm; R120 40 mm.
Ap/V 381, 400 and 401	No valid thickness for any profile mode or supported fire duration, including where underlying tables contain a value.
R90 / 747 °C / Ap/V 379; R120 / 664 °C / Ap/V 17	Unavailable reference cells remain unsupported despite Ap/V being within the maximum.

These are fixtures from the reference data/tests, not permission to generalize thickness between inputs. Exhaustive four-field resolution and row-order independence complement these selected examples.

10.2 Integrated acceptance scenarios

Scenario	Requirements covered	Pass condition
Standard and custom workflow	UX-001; FR-001–FR-007; DR-001–DR-004	Complete both direct and geometry custom modes and standard examples in each language. Missing/unsupported inputs do not produce valid-looking results.
Upward Ap/V rounding	FR-002–FR-003; DR-002–DR-004; UX-010	Exercise all six decimal cases above through direct and geometry paths for R30/R60/R90/R120 in both languages. Exact integers remain unchanged; a rounded value above 380 is rejected before lookup. Missing cells at otherwise permitted rounded values remain unsupported.
Independent project editing	FR-009–FR-014; UX-008–UX-009	Save standard and two differently named custom calculations, switch repeatedly, update only one, add another and cancel/confirm deletion and New Project. Counts, names and unaffected entries remain correct.
Identical profiles using total length	FR-009; FR-019; DR-006	Document ten identical HEB100 members of 6 m as one calculation with 60 m total length and the same fire duration, design temperature, cladding and protected sides. At unchanged waste/build-up, quantities are ten times those for 6 m before display rounding. No profile-count field, extra count multiplier or duplicate calculations are needed.
Production project-file round trip	FR-015–FR-017; EX-007; DR-011	Create and save a project in the production solution, publish newer calculation data, and reopen the project with its original results preserved. Explicitly update one entry; save/reopen and verify changed and unchanged values. Invalid files produce clear errors.
Length and waste validation	FR-019–FR-020; DR-006; UX-004; UX-010	Verify defaults of 6.0 m and 10%. Accept length 0.5 m, waste 0%, decimal waste 12.5% and waste 150% with valid quantities. Reject length 0 or negative, negative waste, blank required input and nonnumeric text with clear localized messages; none silently becomes zero or a valid zero-quantity estimate. Test both comma and period as decimal input separators in both languages.
Current versus saved material results	FR-018–FR-022; DR-005–DR-006	Change length from 6 m to 12 m, waste and build-up. Current quantities refresh; saved project quantities change only after Update Calculation.
Repeated products and CO_2	FR-023; FR-026; DR-007–DR-009	Compare single/repeated board layers and identical/different clamps at 0% and 10% waste. No quantity or emissions are lost through aggregation. Repeat after switching languages.
Environmental basis	FR-025–FR-026; DR-008	Verify that factors supplied from Knauf's product database correspond to product EPDs for A1–A3/C3–C4. Calculations include applicable waste and use the defined bases and exclusions. Excluded screws, staples/clamps and other mapped products remain in material lists without contributing to CO_2.
First-download metadata	FR-008; EX-001–EX-002; EX-005; EX-010; UX-007	Type, replace or clear metadata and immediately click the relevant download without a preliminary field commit. Inspect the first actual PDF or ZIP-contained PDF in both languages, with cold and existing caches. It contains the latest accepted metadata.
PDF date and description	EX-008–EX-009	Exercise UTC midnight, both DST transitions, multiple report pages and later download of unchanged files. Verify two-line bounds/ellipsis for sentences, LF/CRLF, long words and literal special characters in both templates/report types.

Scenario	Requirements covered	Pass condition
Material Excel and ZIP consistency	EX-003–EX-006; FR-022; UX-005	Standalone/project workbooks reconcile to their defined snapshots; the ZIP report matches the project basis. Simulated missing external documents are identified, and repeated links are deduplicated.
Approved export separation	EX-001–EX-005; EX-007	Individual and combined PDFs contain dimensioning, with no material tables or CO_2 schedules. Individual, per-calculation and combined Excel material lists exclude CO_2/EPD/datasheet-link columns. ZIP contains documentation and available supporting files without automatically including editable .fireboard data, which remains a separate save/reopen output. Do not treat the prototype's summed per-metre rates as a project-wide rate or introduce an alternative aggregation formula or additional export type.
Complete localization	UX-002–UX-006; UX-010; DR-009	Audit all application-controlled screens, errors, confirmations, notes, worksheet names and descriptive filenames in both languages. EPIM/PRV descriptions and units match the designated language fields, decimals use the selected convention, and identifiers/user names remain unchanged.
Self-service dataset publication	DR-012–DR-013; DR-016	An authorized Local Data Owner uploads PRV and all related files, reviews validation and publishes a valid update without Knauf Digital intervention or per-update approval. New calculations use the published dataset.
Mandatory PRV content	UX-003; DR-013–DR-014	Reject malformed files, missing mandatory columns, missing Danish or English descriptions and missing required units. Each report identifies the affected file, product and field and the correction needed. A corrected dataset passes validation.
Product identifiers and relationships	DR-009–DR-010; DR-014	Reject missing or conflicting stable product identifiers and broken references between products, material/component rules and calculation datasets. Valid language changes retain identity, units' numeric meaning, quantities and results.
Active production-data protection	DR-015; FR-016; DR-011	Establish active dataset A. Fail an upload and validation of B and verify A remains fully operational and unchanged. Interrupt publication and verify users never see mixed A/B data. Successfully publish valid B; new calculations use B while saved results remain unchanged until explicit update.
Numeric precision and decimal separators	UX-004; FR-019–FR-020; DR-006–DR-008	Show all material quantities and CO_2 results with exactly two decimals, including trailing zeros, using comma in Danish and point in English. Accept both 0,5 and 0.5 as length, and both 12,5 and 12.5 as waste, in either language. Include multiple rows with nonzero third decimals; verify totals are calculated from unrounded values, then displayed with two decimals. Ap/V still follows its separate upward-rounding rule.
EPIM product presentation	UX-003; UX-006; DR-009	Use known PRV products to verify correct Danish/English descriptions and units in screens and applicable exports. Switching language does not translate identifiers or alter quantities or calculation results.
Ownership and access control	DR-012; DR-016	Authorized Local Data Owners can administer and publish data; unauthorized users cannot. Verify that procedures assign dataset maintenance/publication to Knauf Commercial and application security, troubleshooting and availability to Knauf Digital, without a Digital approval gate for each update.

Browser-level tests are necessary for immediate-download metadata, destructive-action confirmations and navigation. Tests must not accidentally pre-commit a field before exercising the download. Network/server delay and repeated

edit/download sequences should be represented in acceptance execution; exact supported browser/device coverage is OQ-11.

The preceding implementation validation of the prototype version supplied in the ZIP reported **390 automated tests passed**, including 132 maximum-Ap/V cases and 89 PDF date/description cases. The suite was not rerun for this documentation-only revision. Existing tests do not establish compliance with the newly approved upward rounding and length/waste validation, or with the confirmation, complete localization and immediate-download target requirements.

11. Prototype Reference and Known Limitations

11.1 Baseline and traceability

Knauf Digital receives the prototype source code as a ZIP copy. References below identify individual source, data, test and documentation files within that copy; source and data links are pinned to prototype commit 546a5effb3226b1beaaba82f1d640f9d1ab3f4f5. The existing docs/functional-specification.md is preserved unchanged. This specification is a separate requirements document, not a revision of that earlier as-built record. The prototype source ZIP is a handover artefact, distinct from the user-generated project documentation ZIP described in EX-005.

Reference area	Relevant sources and tests
Workflow, profile/cladding selection, metadata and download orchestration	app.py: step navigation around 3314; final lookup around 4195–4370; sidebar downloads around 3107–3258; metadata/current PDF around 4495–4578.
Ap/V maximum and supported-result validation	validation_helpers.py, validate_fireboard_lookup; test_apv_maximum.py.
Category-specific lookups	app.py, final Ap/V and material-rate filters; test_profile_category_lookup.py.
Custom geometry and restoration	geometry_helpers.py, custom_profile_validation.py; test_geometry_helpers.py, test_custom_profile_state.py.
Materials, actual identity and CO_2	material_helpers.py, deduplicate_materials, build_material_row, generate_materials; test_material_co2.py, test_screw_clamp_logic.py.
Saved calculations, input settings and snapshots	calculation_state.py; test_calculation_state.py.
Local project files	project_io.py, export_project_state, parse_project_file, import_project_state; test_project_io.py.
PDF output and approved layout	pdf_generator.py, draw_project_description, generate_single_pdf, generate_complete_pdf; Danish template, English template; test_pdf_report_fields.py.
Excel, combined materials and ZIP	export_helpers.py, project_package.py; test_excel_export_styling.py, test_project_package.py.
Localization	translations.py, catalogue description fields, material/export helpers and application-controlled text in app.py.

11.2 Confirmed prototype gaps and boundaries

Topic	Current evidence	Target interpretation
Immediate-download metadata	The description is copied into project metadata after sidebar downloads are processed. Individual PDF bytes are prepared before the click; pending edits can arrive too late for the first download.	UX-007 and EX-010 are approved gaps to close. A framework-specific extra Apply step is not an approved target solution.
Destructive actions	Deletion and New Project execute without confirmation.	Confirmation is required by FR-012, FR-014 and UX-008.
Localization and product content	The prototype uses a local catalogue and description substitutions; some messages, worksheets and filenames remain Danish.	Implement UX-002–UX-006. EPIM centrally maintains Danish/English descriptions and units, supplied via PRV. Mandatory missing descriptions or units prevent publication under DR-014.
Language-independent identity	Combined materials use displayed article numbers, falling back to descriptions; saved snapshots can retain earlier language text.	Stable product identity must govern aggregation. EPIM/PRV relationships use stable identifiers validated before publication; display descriptions must not determine calculation logic.
Historical snapshots	Opening preserves old results; inputs and saved quantities may differ until Update Calculation.	Preservation and explicit updates are approved behavior. Clear explanation is required; automatic recalculation is not.
Custom quantities	No custom-profile material or CO_2 result is calculated. Some project notes have zero numeric placeholders.	Do not invent quantities or represent unavailable estimates as measured zero. Expansion of custom quantities is unapproved.
Numeric input handling	Direct Ap/V is truncated, geometry Ap/V rounded to nearest; unparseable length/waste can become zero and negative inputs are not comprehensively constrained.	Production must round positive Ap/V upward, reject rounded values above 380, accept decimal length > 0 and waste ≥ 0 without a separate upper waste limit, use defaults of 6.0 m/10%, and explain invalid input without silently substituting zero. Material quantity formulas are implemented; input validation requires improvement. Accept both decimal separators and apply exactly two decimals to material/CO_2 display, with unrounded totals, under UX-004.
Identical profiles	The prototype supports total profile length and suppresses an exact repeat of the last saved calculation.	One calculation may cover multiple profiles with identical engineering conditions using their combined length. A profile-count field and duplicate-calculation feature are not required.
Environmental data and calculations	Factors originate from Knauf's existing product database, based on product EPDs covering A1–A3 and C3–C4. Product-specific logic and exclusions are implemented and tested.	Preserve the calculation bases and defined BR18 exclusions; excluded components remain in material lists. Apply UX-004 for presentation and total precision.
Export boundaries	PDFs omit material/CO_2 schedules; Excel omits CO_2/document links; ZIP excludes editable project files. Combined per-metre rates are summed.	Preserve the approved separation and existing content, including both project material-list options. The summed per-metre-rate column is prototype behavior, not a meaningful project-wide rate; no replacement aggregation formula or new export type is approved.
Data administration	The prototype uses local data files; no self-service EPIM/PRV upload, validation or publication portal is implemented.	Implement DR-012–DR-016 with Local Data Owner publication, automatic validation and protection of active production data.

Topic	Current evidence	Target interpretation
Local persistence and external documents	No project database/accounts are implemented. Supporting documents depend on external URLs and can be omitted with a README when unavailable.	Local save/reopen remains required. Knauf Digital decides the production platform; external-document availability is not guaranteed by the application.

The current baseline already includes the category-specific Ap/V/material lookup correction, full aggregated Fireboard-area CO_2 correction, distinct clamp articles, custom-profile restoration/reset correction, enforced integer Ap/V maximum, and Copenhagen generation dates/two-line PDF descriptions. Earlier as-built statements that these last two PDF behaviors or the 380 limit are missing are obsolete for this baseline.

12. Open Questions and Responsibilities

The scope and users, production project persistence, EPIM/PRV workflow, mandatory validation, self-service data ownership, numerical precision and current-versus-saved calculation semantics are approved. OQ-01 and OQ-05 are removed; OQ-08 is resolved by UX-003 and DR-014, and OQ-09 by UX-004 and DR-002/DR-006–DR-008. Gaps in question numbering preserve earlier references. Remaining questions concern implementation, operational arrangements and acceptance, without reopening these decisions.

Responsibility area	Responsible party
Functional scope and product requirements	Knauf Commercial / designated product owner
Maintenance, upload and publication of calculation and product datasets	Knauf Commercial – Local Data Owner, independently through the portal
Central product descriptions and units	EPIM; delivered through PRV with each Fireboard dataset update
Engineering reference content and fire-protection assumptions	Relevant Knauf technical/R&D owner supplies specialist input to the Local Data Owner
Product-EPD environmental content and defined exclusions	Relevant Knauf sustainability/data owner supplies specialist input to the Local Data Owner
Application, technical architecture, hosting, security, maintenance, troubleshooting and operational availability	Knauf Digital
Functional production acceptance	Knauf Commercial and Knauf Digital jointly

12.1 Reference and user-presentation acceptance

ID	Remaining decision or evidence	Responsibility and boundary
OQ-04	Identify the exact engineering reference release and supporting geometry guidance, component rules and product-EPD documents to use for production acceptance.	Local Data Owner coordinates the acceptance evidence with the relevant technical/R&D and sustainability/data owners. Factor provenance and A1–A3/C3–C4 scope are approved. This acceptance preparation does not require Knauf Digital approval for individual dataset updates.
OQ-10	Finalize the detailed user presentation of current edits and saved calculation/material results, including changes since an explicit update.	Knauf Digital implements the interaction with Knauf Commercial. The semantics are already approved: individual exports use the current valid calculation; project exports use saved results; latest metadata is accepted at download; saved results change only through explicit review and update. The remaining issue is presentation, not these rules.

12.2 Implementation, operation and acceptance

ID	Remaining decision or evidence	Responsibility and boundary
OQ-11	Agree production acceptance execution and evidence, supported browsers/devices, and operational arrangements.	Knauf Commercial and Knauf Digital agree functional acceptance. Knauf Digital determines technical implementation, hosting, security, maintenance, troubleshooting and availability arrangements. Detailed access-control implementation must enforce DR-016. Browser coverage, service levels and operational procedures remain to be specified; data maintenance and publication ownership are already assigned to the Local Data Owner.

The Local Data Owner uploads, validates and publishes data without manual intervention or per-update approval from Knauf Digital. Technical choices remain with Knauf Digital. No additional functionality or new organizational approval process is implied by the remaining questions.